Forecasting Future Fluorescence Expression from Early Brightfield Organoid Imaging

OrganoidAgent Yichao Forecasting Design

2026-05-10

Contents

1 Goal

The practical goal is not only to translate a same-time brightfield image into a same-time fluorescence image. The real biological goal is to predict later fluorescence expression before the organoid visibly expresses fluorescence.

There are therefore two related tasks:

1. **Same-time translation:** learn $B_t \mapsto F_t$, where B_t is brightfield and F_t is fluorescence at the same time and depth.
2. **Future forecasting:** learn $B_{1:k} \mapsto Y_{>k}$, where $B_{1:k}$ is an early brightfield history before detectable expression and $Y_{>k}$ summarizes future fluorescence expression.

The second task is the target method. Same-time pix2pix remains useful because it validates image registration, channel pairing, segmentation quality, and whether morphology contains immediate fluorescence-related information. It does not by itself solve future prediction.

2 Channel Convention

The current instance-pair segmentation/database pipeline uses:

- $c1$: brightfield input,
- $c0$: fluorescence target.

Some older notes used the opposite wording. Before training any model, verify the channel by opening paired files such as:

```
*_t000_z000_c1.jpg  brightfield
*_t000_z000_c0.jpg  fluorescence
```

3 Which Data Can Be Used

3.1 Find datasets with time information first

Future forecasting requires time-indexed data. A static one-frame image can train same-time translation, but it cannot answer whether early morphology predicts later expression.

The exported Yichao filenames have this form:

```
<series>_<object_name>_tTTT_zZZZ_cC.jpg
```

A folder has time information if at least one object has more than one unique t-index:

$$T_{\text{object}} = |\{t : \exists z, c\ \text{file}(object, t, z, c)\}| > 1. \tag{1}$$

For paired supervised data, the brightfield and fluorescence files must share the same object, time, and depth:

$$(B_{p,t,z}, F_{p,t,z}) = (I_{p,t,z,c=1}, I_{p,t,z,c=0}), \tag{2}$$

where p denotes the position or object folder.

3.2 Real Yichao time-aware inventory

Table 1 summarizes the current usable Yichao folder structure discovered from the grouped JPEG roots. Pair counts are 2D paired planes, not organoid-instance counts.

Table 1: Yichao paired-frame inventory. $T_{\max}$ and $Z_{\max}$ are the maximum number of timepoints and z-planes observed in a position.

Dataset	Main role	Pairs	Positions	$T_{\max}$	$Z_{\max}$	Day tags
Data-Yichao-1	static check only	5	5	1	1	none
Data-Yichao-2	dynamic D2 plus duplicated static images	965	11	16	11	D2
Data-Yichao-3	dynamic monitoring	10019	13	49	32	Day 2–4
Data-Yichao-4	dynamic monitoring	7940	9	49	38	Day 2–3
Data-Yichao-5	dynamic monitoring	10143	12	45	34	D1–D3
Data-Yichao-6	dynamic D4 monitoring	4410	3	45	40	D4
Data-Yichao-7	dynamic D1–D3 monitoring	7706	8	38	39	D1–D3
Data-Yichao-8	dynamic D2–D3 monitoring	12536	18	45	36	D2–D3
Data-Yichao-9	PDO28/Jurkat live imaging	4756	3	116	15	D2–D3
Data-Yichao-10	dynamic monitoring	16605	13	41	46	DF

For the forecasting task, prioritize datasets with clear time courses and repeated positions:

- first: Data-Yichao-3, 4, 5, 7, 8, 10;
- optional after inspection: Data-Yichao-2 dynamic D2 positions and Data-Yichao-6 D4 positions;
- separate biological task: Data-Yichao-9 because it contains PDO28/Jurkat live imaging and T-cell fluorescence.

3.3 Concrete examples for datasets 6–8

The currently processed set $6, 7, 8$ contains exactly the kind of time/depth structure needed for a future model:

- Data-Yichao-6: 3 D4 positions, 45 timepoints, 25–40 z-planes.
- Data-Yichao-7: 8 positions across D1, D2, D3, 31–38 timepoints, 17–39 z-planes.
- Data-Yichao-8: 18 positions across D2 and D3, 32–45 timepoints, 9–36 z-planes.

The most useful forecasting subset is any same-position or same-condition sequence where early frames precede fluorescence onset and later frames contain expression.

4 Recommended Data Representation

4.1 Frame-level pair

The lowest-level pair is:

$$\mathcal{P}_{p,t,z} = (B_{p,t,z}, F_{p,t,z}). \tag{3}$$

This pair is enough for same-time pix2pix, but it is not enough for forecasting.

4.2 Instance-level pair

After segmentation, every organoid instance i has:

$$\mathcal{I}_{i,p,t,z} = \left(B^{\text{crop}}_{i,p,t,z}, F^{\text{crop}}_{i,p,t,z}, M_{i,p,t,z}, m_{i,p,t,z}\right), \tag{4}$$

where M is the mask and m is a vector of morphology features.

For the first forecasting model, keep only complete instances:

$$\text{complete}(i,p,t,z) = \mathbf{1}[\text{is\_edge\_padded} = 0]. \tag{5}$$

4.3 Track-level sample

The correct forecasting unit is an organoid track:

$$\mathcal{T}_j = \left\{\left(B^{\text{crop}}_{j,t}, F^{\text{crop}}_{j,t}, M_{j,t}, m_{j,t}\right)\right\}_{t=1}^{T_j}. \tag{6}$$

Here j indexes a physical organoid tracked through time. In practice, $B_{j,t}$ should be a 2D representation per timepoint, not a full z-stack at first. Good first choices are:

- central z-plane,
- best-focus z-plane,
- max or mean projection,
- one stable z-plane if the z-index is biologically meaningful.

5 Pipeline Flow

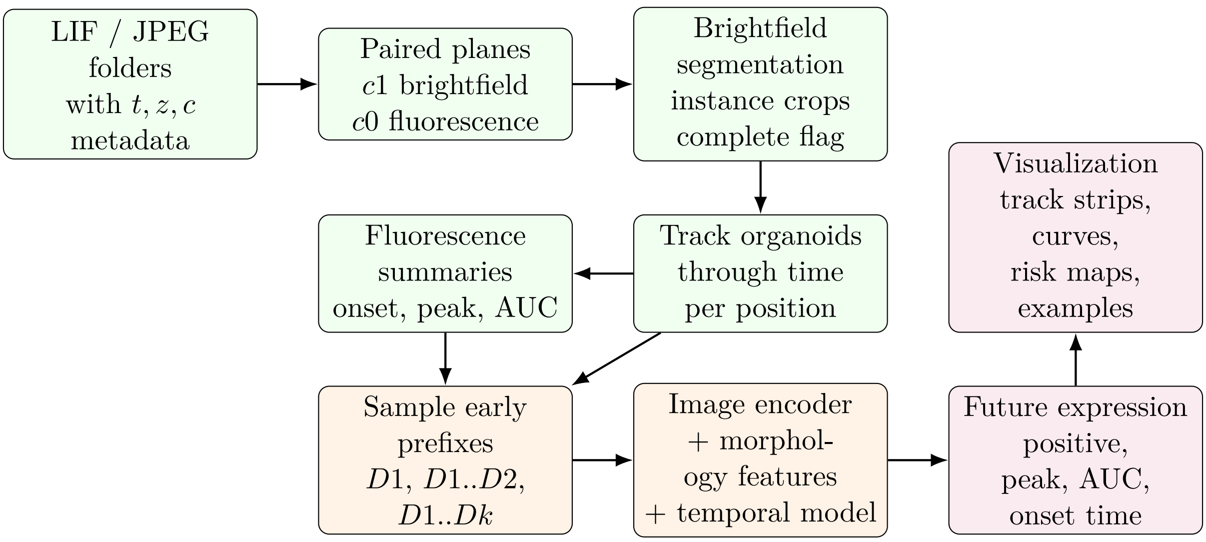

Figure 1: Forecasting pipeline from real Yichao time-indexed image folders to future expression predictions.

6 Tracking

6.1 Simple first tracker

For every position p, timepoint t, and chosen z representation, the segmentation pipeline produces instances $a \in A_t$ and $b \in A_{t+1}$. Link them by minimizing a cost:

$$C(a,b) = \lambda_d \frac{\|c_a - c_b\|_2}{r_a + r_b + \epsilon} \lambda_o \left(1 - \mathrm{IoU}(M_a, M_b)\right) \lambda_s \left|\log \frac{A_a + \epsilon}{A_b + \epsilon}\right|, \tag{7}$$

where c is centroid, r is equivalent radius, M is mask, and A is mask area.

Use Hungarian matching or greedy nearest-neighbor matching under a maximum cost threshold:

$$(a,b) \text{ is linked} \quad \Longleftrightarrow \quad b = \arg\min_{b' \in A_{t+1}} C(a,b') \;\text{ and }\; C(a,b) < \tau_{\text{track}}. \tag{8}$$

Break a track if no confident match exists. Keep tracks satisfying:

$$T_j^{\text{early}} \geq T_{\min}^{\text{early}}, \qquad T_j^{\text{future}} \geq T_{\min}^{\text{future}}, \qquad \text{complete}(j,t) = 1 \text{ for most frames.} \tag{9}$$

6.2 Why not full 3D tracking first

Full 3D tracking is possible but not the best first implementation. The early goal is to learn whether brightfield morphology predicts later expression. A stable 2D projection or best-focus plane is easier to validate and less sensitive to z-plane noise.

7 Fluorescence Signal and Onset

7.1 Background-corrected fluorescence

For each track j and time t, define the organoid mask $M_{j,t}$ and a local background ring $R_{j,t}$ around the organoid. A robust fluorescence score is:

$$S_{j,t} = \text{quantile}_{0.90}\left(F_t(x) : x \in M_{j,t}\right) - \text{median}\left(F_t(x) : x \in R_{j,t}\right). \tag{10}$$

Also compute:

$$\bar{S}_{j,t} = \text{mean}\left(F_t(x) : x \in M_{j,t}\right) - \text{median}\left(F_t(x) : x \in R_{j,t}\right), \tag{11}$$

$$q_{j,t} = \frac{1}{|M_{j,t}|} \sum_{x \in M_{j,t}} \mathbf{1}[F_t(x) > \theta_t], \tag{12}$$

$$P_j = \max_t S_{j,t}, \tag{13}$$

$$AUC_j = \sum_{t=1}^{T_j} \max(S_{j,t}, 0)\Delta t. \tag{14}$$

7.2 Onset time

Let a positive threshold be:

$$\theta_j = \max\left(Q_{0.95}^{\text{negative}}, \text{median}(R_{j,t}) + 3\,\text{MAD}(R_{j,t})\right). \tag{15}$$

Define onset as the first sustained positive time:

$$\tau_j = \min\left\{t : \; S_{j,t} > \theta_j \text{ and } S_{j,t+1} > \theta_j\right\}. \tag{16}$$

If no such t exists, the track is treated as censored or negative:

$$y_j^{\text{pos}} = 0, \qquad \tau_j = \varnothing. \tag{17}$$

7.3 Guard against leakage

The input prefix must end before visible fluorescence:

$$k < \tau_j - g, \tag{18}$$

where $g \in \{1, 2\}$ is a guard interval. This prevents the model from using weak fluorescence that has already started.

8 Forecasting Model

8.1 Input

For a track j, choose a prefix length k. The input is:

$$X_{j,1:k} = \left\{ B^{\text{crop}}_{j,1}, m_{j,1}, \Delta t_1;\ \ldots;\ B^{\text{crop}}_{j,k}, m_{j,k}, \Delta t_k \right\}. \tag{19}$$

8.2 Encoder

Use one image encoder shared across time:

$$e_{j,t} = \phi_\theta \left(B^{\text{crop}}_{j,t} \right). \tag{20}$$

Concatenate morphology features and elapsed time:

$$x_{j,t} = [e_{j,t}, m_{j,t}, \Delta t_t]. \tag{21}$$

Recommended morphology features include area, diameter, circularity, aspect ratio, lumen or hollow fraction, edge sharpness, texture heterogeneity, growth rate, change in circularity, and budding/protrusion score.

8.3 Temporal aggregation

Use a small recurrent or attention model:

$$h_{j,k} = \text{GRU}_\psi(x_{j,1}, x_{j,2}, \ldots, x_{j,k}). \tag{22}$$

Prediction heads are:

$$\hat{y}^{\text{pos}}_j = \sigma(w^\top_{\text{pos}} h_{j,k}), \tag{23}$$

$$\hat{P}_j = f_{\text{peak}}(h_{j,k}), \tag{24}$$

$$\widehat{AUC}_j = f_{\text{auc}}(h_{j,k}), \tag{25}$$

$$\hat{\tau}_j = f_{\text{onset}}(h_{j,k}). \tag{26}$$

9 Training Objective

Use one variable-prefix model instead of one model per day. During training, sample k randomly from valid pre-expression prefixes:

$$k \sim \text{Uniform}\{1, \ldots, \tau_j - g - 1\}. \tag{27}$$

The multi-task loss is:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}(y^{\text{pos}}_j, \hat{y}^{\text{pos}}_j) + \alpha\, \text{SmoothL1} \left(\log(1 + P_j), \hat{P}_j \right) + \beta\, \text{SmoothL1} \left(\log(1 + AUC_j), \widehat{AUC}_j \right) + \gamma \mathcal{L}_{\text{ordinal}}(\tau_j, \hat{\tau} \tag{28}$$

This single model learns:

$$M(D1), \quad M(D1, D2), \quad \ldots, \quad M(D1, \ldots, Dk) \tag{29}$$

because the prefix length is varied during training.

10 How to Choose the Shortest Useful Prefix

Evaluate the model by prefix length. Let $S(k)$ be the validation score at prefix length k, for example AUROC for future-positive classification or Spearman correlation for future-peak prediction.

The recommended decision rule is:

$$k^\star = \min \left\{ k : \ S(k) \geq 0.95 \max_{\ell} S(\ell) \right\}. \tag{30}$$

Interpretation:

- If $k^\star = 1$, earliest brightfield morphology is already informative.
- If $k^\star = 2$ or 3, growth dynamics before expression are important.
- If performance only improves after expression onset, the task is leaking and must be redefined.

11 Evaluation

Splits must be track-level and preferably position-level:

$$\mathcal{T}_{\text{train}} \cap \mathcal{T}_{\text{test}} = \varnothing, \qquad \mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{test}} = \varnothing \text{ for the final test.} \tag{31}$$

Do not split adjacent frames from the same position across train and test.

Recommended metrics:

- future positive: AUROC, AUPRC, F1, sensitivity at fixed specificity;
- future peak and AUC: Spearman, Pearson, MAE, R^2;
- onset time: mean absolute error in hours or days, early/late confusion matrix;
- calibration: Brier score and reliability curve.

Always compare against simple baselines:

- current organoid area only;
- growth rate only;
- morphology-feature random forest or XGBoost;
- dataset/position average;
- same-time pix2pix encoder features without temporal history.

12 Visualization

12.1 What to visualize

The model should be inspected visually at every stage:

1. **Pair QC grid:** brightfield and fluorescence crops side by side for each dataset.
2. **Track strip:** brightfield crops over time, fluorescence crops over time, and segmentation overlays.

3. **Onset curve:** $S_{j,t}$, threshold θ_j, predicted risk, and detected onset τ_j.
4. **Prefix performance:** validation score $S(k)$ versus input prefix length.
5. **Interpretability:** Grad-CAM or occlusion map on brightfield frames, plus feature importance for morphology features.

12.2 Existing pair-grid demo

The repository already contains a QC pair grid produced from the resized instance-pair database. If present, it can be included directly:

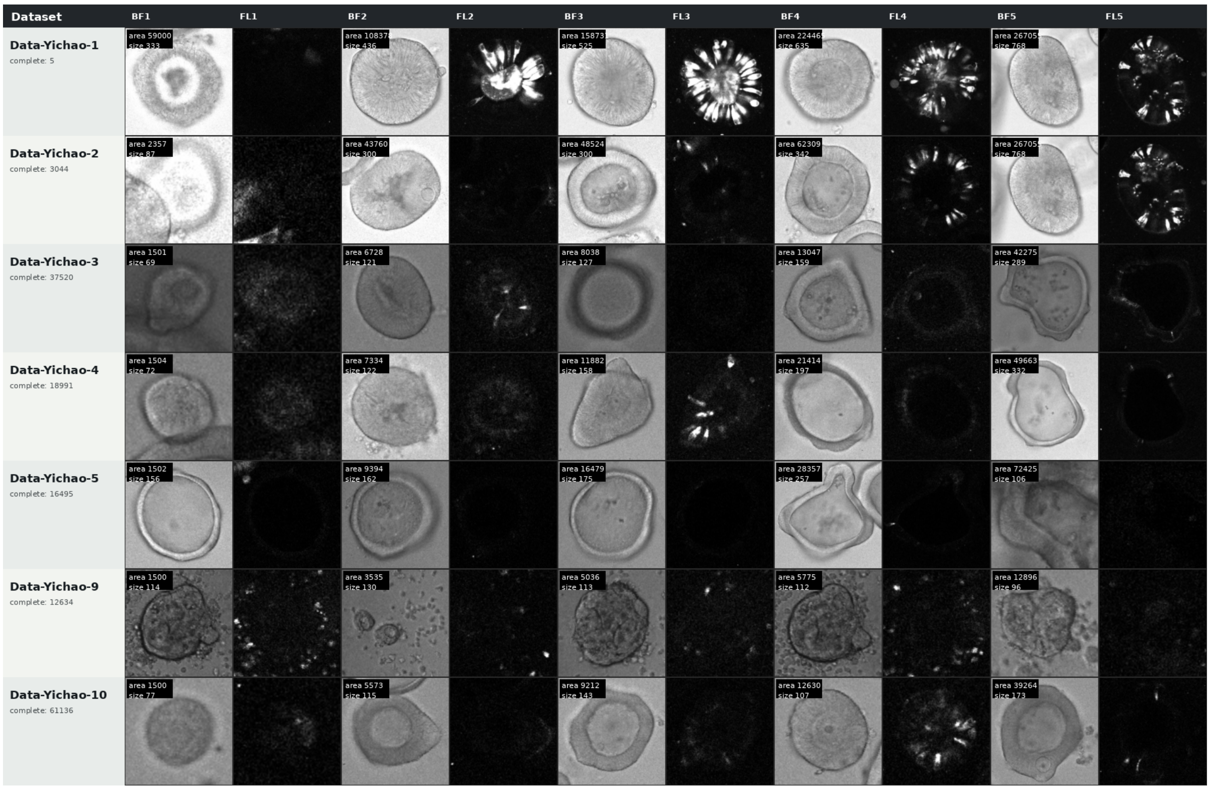

Figure 2: Existing complete-organoid instance-pair visualization. Each dataset row shows paired brightfield/fluorescence crops for complete non-edge-padded organoids. This is a QC visualization, not yet a temporal forecast visualization.

12.3 Forecasting curve

For each tracked organoid, plot:

$$\left\{(t, S_{j,t}), (t, \hat{y}_{j,t}^{\text{pos}})\right\}_{t=1}^{T_j}. \tag{32}$$

Overlay:

- fluorescence threshold θ_j,
- detected onset τ_j,
- prefix endpoint k,
- future target window.

This plot is the most important debugging visualization because it reveals whether the model is predicting before expression or merely reacting after expression has already begun.

13 Concrete Demo Pipeline

13.1 Step 1: select time-aware data

Use only folders where $T > 1$. In the current repository, the relevant grouped roots include:

```
Data-Yichao-3/N39_TriRep_DF_jpeg_all_by_position
Data-Yichao-4/N39_TriRep_DF_2_jpeg_all_by_position
Data-Yichao-5/N39_TriRep_DF_3_jpeg_all_by_position
Data-Yichao-6/N39_TriRep_4_jpeg_all_by_position
Data-Yichao-7/N39_TriRep_5_jpeg_all_by_position
Data-Yichao-8/N39_TriRep_DF_6_jpeg_all_by_position
Data-Yichao-10/N39_TriRep_DF_7_jpeg_all_by_position
```

13.2 Step 2: build instance database

The current segmentation pipeline writes:

```
analysis-outputs/yichao_instance_pairs/images/
analysis-outputs/yichao_instance_pairs/instances/
analysis-outputs/yichao_instance_pairs/database/instance_pairs.sqlite
```

Use the database to filter complete organoids:

```
SELECT *
FROM instances
WHERE CAST(is_edge_padded AS INTEGER) = 0;
```

13.3 Step 3: create tracks

Create a new track table:

```
tracks(track_id, dataset, object_name, time_index, z_index,
       instance_id, centroid_x, centroid_y, area_px,
       is_edge_padded, fluorescence_score, track_quality)
```

For the first implementation, build one 2D representation per timepoint using a best-focus or projection rule.

13.4 Step 4: create future labels

Create a track-label table:

```
track_labels(track_id, future_positive, onset_time_index,
             future_peak, future_auc, censor_flag)
```

13.5 Step 5: train and visualize

Train the variable-prefix model and produce:

- prefix-length performance plot;
- prediction examples for high-confidence positive and negative tracks;
- onset curves with predicted risk;
- morphology feature importance.

14 Future Image Prediction

After scalar forecasting works, add an image decoder:

$$\hat{F}^{\text{crop}}_{j,t+h} = D_\omega(h_{j,k}, h), \tag{33}$$

where h is the forecast horizon. Train with mask-restricted image losses:

$$\mathcal{L}_{\text{image}} = \left\| M_{j,t+h} \odot \left(\hat{F}^{\text{crop}}_{j,t+h} - F^{\text{crop}}_{j,t+h} \right) \right\|_1 \;+\; \eta\,\mathcal{L}_{\text{SSIM}} \;+\; \rho\,\mathcal{L}_{\text{Dice}}. \tag{34}$$

Do not start with this image task. It is harder because future organoid geometry is not fixed. The scalar future-expression model should be the first scientifically interpretable result.

15 Main Risks and Controls

- **Tracking risk:** if same-organoid tracking is unreliable, switch to position-level forecasting.
- **Leakage risk:** do not allow weak fluorescence in the input prefix; use an onset guard.
- **Split risk:** split by position or LIF file, not by individual planes.
- **Artifact risk:** debris can have strong fluorescence; use morphology and complete-organoid masks to reject non-organoid objects.
- **Class imbalance:** if positive tracks are rare, report AUPRC and calibrate thresholds.

16 Minimal First Experiment

The first robust experiment should be:

- data: complete tracked organoids from time-aware Yichao datasets;
- input: early brightfield crop sequence before onset;
- target: future positive, $\log(1 + \text{peak})$, $\log(1 + \text{AUC})$;
- model: CNN encoder plus morphology features plus GRU;
- split: held-out positions;
- decision: find the shortest prefix $k^\star$ with near-best validation performance.

This directly answers whether early brightfield morphology can predict later differentiation-related fluorescence expression.